AWG Wire Size Chart

Every standard size from 18 AWG to 1000 kcmil, with its diameter, its area in circular mils and square millimetres, and its resistance in copper and aluminum.

In the American Wire Gauge system a smaller number is a larger wire, and every three sizes the area doubles. A 10 AWG conductor has about twice the copper of a 13 AWG and half the resistance; go up six sizes and the diameter doubles. Above 4/0 the gauge numbers stop and sizes are given directly by area, in thousands of circular mils (kcmil).

AWG and kcmil wire sizes in metric units

Size	Diameter (mm)	Area (mm²)	Copper (Ω/km)	Aluminum (Ω/km)	Next IEC size (mm²)
18 AWG	1.02	0.821	26.1	42.9	1
16 AWG	1.29	1.31	16.4	27.0	1.5
14 AWG	1.63	2.08	10.3	16.9	2.5
12 AWG	2.05	3.31	6.48	10.7	4
10 AWG	2.59	5.26	4.08	6.70	6
8 AWG	3.26	8.37	2.56	4.21	10
6 AWG	4.11	13.3	1.61	2.65	16
4 AWG	5.19	21.1	1.01	1.67	25
3 AWG	5.83	26.7	0.804	1.32	35
2 AWG	6.54	33.6	0.638	1.05	35
1 AWG	7.35	42.4	0.506	0.831	50
1/0 AWG	8.25	53.5	0.401	0.659	70
2/0 AWG	9.27	67.4	0.318	0.523	70
3/0 AWG	10.40	85.0	0.252	0.415	95
4/0 AWG	11.68	107	0.200	0.329	120
250 kcmil	12.70	127	0.169	0.278	150
300 kcmil	13.91	152	0.141	0.232	185
350 kcmil	15.03	177	0.121	0.199	185
400 kcmil	16.06	203	0.106	0.174	240
500 kcmil	17.96	253	0.085	0.139	300
600 kcmil	19.67	304	0.071	0.116	400
750 kcmil	22.00	380	0.056	0.093	400
1000 kcmil	25.40	507	0.042	0.070	630

Diameter is that of a solid round conductor of the stated area. The last column is the next standard IEC conductor size at or above the AWG area, which is the usual choice when a metric cable replaces an AWG one. Resistance is DC, at about 75°C.

How the values are calculated

$$\text{Area (mm}^2\text{)} = \pi / 4 \times d^2, \text{ with } d \text{ in mm}$$

$$R\ (\Omega/\text{km}) = \rho \times 1000 / A, \text{ with } \rho = 0.0214 \text{ for copper and } 0.0352 \text{ for aluminum, in } \Omega\cdot\text{mm}^2/\text{m}$$

Every figure in the chart comes from the formulas above. Nothing is copied from a published table, so any cell can be checked by hand.

What this chart does not tell you

This chart gives geometry and resistance. It does not give ampacity, which depends on insulation temperature rating, how many conductors share a raceway, the ambient temperature and the terminations, and has to come from the adopted code. Resistance is DC; on large conductors carrying AC, skin effect raises the effective figure.

This is a reference, not a design. It is here to give you the right order of magnitude and a figure to check against. It does not replace the adopted code or a calculation by a qualified engineer.

From voltfield.org/charts/awg-wire-size-chart.html, where the notes on reading this chart and a calculator for other cases are. Reviewed November 11, 2026. Free to print, copy and share.